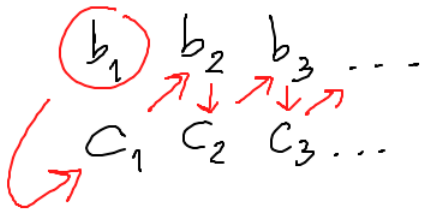
$B = \{b_1, b_2, b_3, \dots\}$

$C = \{c_1, c_2, c_3, \dots\}$

$\mathbb{N}$	$B \cup C$
1	b_1
2	c_1
3	b_2
4	c_2
5	$\vdots$
$\vdots$	$\vdots$



f

g

$$h(n) = \begin{cases} ? & n \dots \\ ? & n \dots \end{cases}$$

$A_{TM} = \{\langle M, w\rangle \mid M \text{ is a TM that accepts } w\}$ is undecidable.

Given a TM M and a string w. Is $\langle M, w\rangle \in A_{TM}$???

We cannot distinguish whether M accepts w or M does not accept w

---

$E_{TM} = \{\langle M\rangle \mid M \text{ is a TM and } L(M) = \emptyset\}$

Assume that E_{TM} is decidable. We know whether $\langle M'\rangle \in E_{TM}$ or $\langle M'\rangle \notin E_{TM}$ for a TM M'

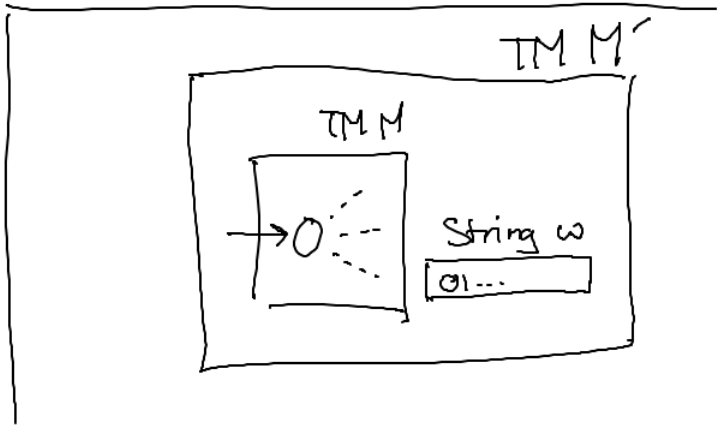
Assume that TM R decides E_{TM}

Run TM R on input $\langle M'\rangle$

- R accepts $\langle M'\rangle$, $\langle M'\rangle \in L(R) = E_{TM}$, $L(M') = \emptyset$
- R rejects $\langle M'\rangle$, $\langle M'\rangle \notin L(R) = E_{TM}$, $L(M') \neq \emptyset$

Can R help us decide A_{TM}?

- Yes, but not directly



Given TM M and a string w

M' = "On input x:
1. Run M on input w'
2. If M accepts w, accept
   If M rejects w, reject"

If M accepts w, $L(M') = \Sigma^* \neq \emptyset$

If M rejects w, $L(M') = \emptyset$

If M loops indefinitely on w, $L(M') = \emptyset$

Run R on input $\langle M' \rangle$

- If R accepts $\langle M' \rangle$, $\langle M' \rangle \in L(R) = E_{TM}$, $L(M') = \emptyset$, M does not accept w, $\langle M, w \rangle \notin A_{TM}$
- If R rejects $\langle M' \rangle$, $\langle M' \rangle \notin L(R) = E_{TM}$, $L(M') \neq \emptyset$, M accepts w, $\langle M, w \rangle \in A_{TM}$

```
M (String s)
{
  ⋮
}
String w = "...";

M'(string s)
{
   M(w);
   ⋮
}
```

$E_{TM} = \{ \ldots \}$

Assume that E_{TM} is decidable by TM R

Construct TM S to decide A_{TM}

S = "On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct TM M' as discussed
2. Run TM R on input $\langle M' \rangle$
3. If R accepts $\langle M' \rangle$, reject
   If R rejects $\langle M' \rangle$, accept"

1. If $\langle M, w \rangle \in A_{TM}$, S will accept $\langle M, w \rangle$
2. If $\langle M, w \rangle \notin A_{TM}$, S will reject $\langle M, w \rangle$

$REGULAR_{TM} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is regular}\}$

Assume that $REGULAR_{TM}$ is decidable. There exists a TM R that decides regular TM.

Option #1:

- If M accepts w, L(M') is regular Σ^*
- If M does not accept w, L(M') is not regular. $\{0^n 1^n \mid n \geq 0\}$

M' = "On input x:

1. if x is in the form of $0^n 1^n$ for $n \geq 0$, accept
2. If x is not in the form of $0^n 1^n$, run M on input w.
3. If M accepts w, accept
   If M rejects w, reject "

If M accepts w,
$L(M') = \Sigma^*$ regular

If M rejects w,
$L(M') = \{0^n 1^n \mid n \geq 0\}$

If M loops indefinitely on w
$L(M') = \{0^n 1^n \mid n \geq 0\}$
not regular

Option #2:

- If M accepts w, L(M') is not regular $\{0^n1^n \mid n \geq 0\}$
- If M does not accept w, L(M') is regular $\emptyset$

S = "On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct TM M'' as follows:

   M'' = "On input x:

   1. Run M on input w
   2. If M accepts w and x is in the form of 0^n1^n for $n \geq 0$, accept
   3. reject"

   TM M' from previous whiteboard

2. Run R on input $\langle M'' \rangle$ (M')
3. If R accepts $\langle M'' \rangle$ (M'), reject (accept)
   If R rejects $\langle M'' \rangle$ (M'), accept (reject) "

$EQ_{TM} = \{\langle M_1, M_2\rangle \mid M_1 \text{ and } M_2 \text{ are TMs and } L(M_1) = L(M_2)\}$

Assume that EQ_{TM} is decidable by TM R.

Construct TM S to decide A_{TM}

S = "On input $\langle M, w\rangle$ where ....

1. Construct TM M_1 as follows

   M_1 = "On input x:
   1. accept" $\}\ L(M_1) = \Sigma^*$

2. Construct TM M_2

   M_2 = "On input x:
   1. Run M on input w
   2. If M accepts w, accept
      If M rejects w, reject"

3. Run R on input $\langle M_1, M_2\rangle$

4. If R accepts $\langle M_1, M_2\rangle$, accept
   If R rejects $\langle M_1, M_2\rangle$, reject"